GEOMETRIC REDUCEDNESS AT A POINT TESTED OVER PURELY INSEPARABLE EXTENSIONS AND OVER THE PERFECT CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and covers all four conditions through the two cited criteria; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-check-only-finite-inseparable-extensions is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A purely inseparable base change induces a homeomorphism and identifies the local ring with the tensor product. Algebra, Lemma algebra-lemma-p-ring-map, invoked at proof lines 479-484. Supplies the uniqueness of the lifted point and the identification of the local ring.
- Pointwise criterion for geometric reducedness of a scheme at a point. Same chapter, Lemma lemma-geometrically-reduced-at-point, invoked at proof lines 485-486. Supplies the passage between the scheme statement and the local rings.
- Geometric reducedness tested over finite purely inseparable extensions. Algebra, Lemma algebra-lemma-geometrically-reduced-finite-purely-inseparable-extension, invoked at proof lines 486-487. Supplies the algebraic equivalence.

Source reference. Original lemma 030V: sources/stacks-chow/tags.txt, source lines 3962–3962. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 035W: sources/stacks-varieties/pinned-original-source.tex, source lines 354–354. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a scheme over k. Let $x \in X$. The following are equivalent*

(1) *X is geometrically reduced at x,*
(2) *$\mathcal{O}_{X_{k'},x'}$ is reduced for every finite purely inseparable field extension k' of k and $x' \in X_{k'}$ the unique point lying over x,*
(3) *$\mathcal{O}_{X_{k^{1/p}},x'}$ is reduced for $x' \in X_{k^{1/p}}$ the unique point lying over x, and*
(4) *$\mathcal{O}_{X_{k^{perf}},x'}$ is reduced for $x' \in X_{k^{perf}}$ the unique point lying over x.*

Proof. Note that if k'/k is purely inseparable, then $X_{k'} \to X$ induces a homeomorphism on underlying topological spaces, see Algebra, Lemma 0BRA. Whence the uniqueness of x' lying over x mentioned in the statement. Moreover, in this case $\mathcal{O}_{X_{k'},x'} = \mathcal{O}_{X,x} \otimes_k k'$. Hence the lemma follows from Lemma 035W above and Algebra, Lemma 030V. □